Pattern Recognition

Homework #2 — K-Nearest Neighbors Classification

About the Assignment

The main aim of this assignment is to make you familiar with a traditional classifier by using the K-Nearest Neighbors (KNN) algorithm. The learning objectives of this lab are:

- Learning the K-NN classification method with custom similarity.
- Understanding the idea behind the classification task.

Algorithm Overview

The algorithm for KNN classification can be described as follows:

1. Compute the k nearest neighbors from the training dataset for the test sample to be classified.
2. Classify the test sample to the class having the most training samples among the k nearest neighbors.

Instructions

Step 1 — Download the CIFAR-10 Dataset

Download the CIFAR-10 dataset (Python version) by using the following commands:

```
from keras.datasets import cifar10
(x_train, y_train), (x_test, y_test) = cifar10.load_data()
```

Step 2 — Convert Images to Vector Format

Convert images to vector format using the snippet below. Each image of size 32×32×3 is converted to a 1×3072 vector. The dataset contains 50,000 training vectors and 10,000 test vectors.

```
x_train = x_train.reshape(-1, 3072)
x_test = x_test.reshape(-1, 3072)
```

Step 3 — Implement the KNN Function

Write a function that accepts x_train, y_train, sample_test, and k as input parameters and returns the most similar class name for sample_test. For computing the similarity, use the Custom Similarity measure explained in the lecture notes. The following example code may help you:

```
import numpy as np

x = np.array([-0.2, 0])    # assume x is test sample (x_test)
y = np.array([1, 1])       # assume y is one of train sample
x = x.reshape(1, -1)
y = y.reshape(1, -1)

s = custom_similarity(x, y)
print("similarity:", s)
```

The variable sample_test refers to a test vector from x_test. You can set it as follows:

```
sample_test = x_test[1, :]
```

You will compute similarities between test and train samples, then sort them in descending order (e.g., 100.45, 87.20, 65.40, 40.12, 23.10, 18.10, 11.11, 5.00). Choose the most similar ones based on k = 5.

The test code should look like this:

```
sample_test = x_test[0, :]
k = 5
similar_class_name = knnCustomSimilarity(x_train, y_train, sample_test, k)
print(similar_class_name)
```

Submission

File Naming Convention

Submit your assignment as a ZIP file using the following format:

`No_Name_Surname_HW2.zip`